



Transparent icephobic coatings using bio-based epoxy resin

Xinghua Wu^a, Shunli Zheng^a, Daniel A. Bellido-Aguilar^a, Vadim V. Silberschmidt^b, Zhong Chen^{a,*}

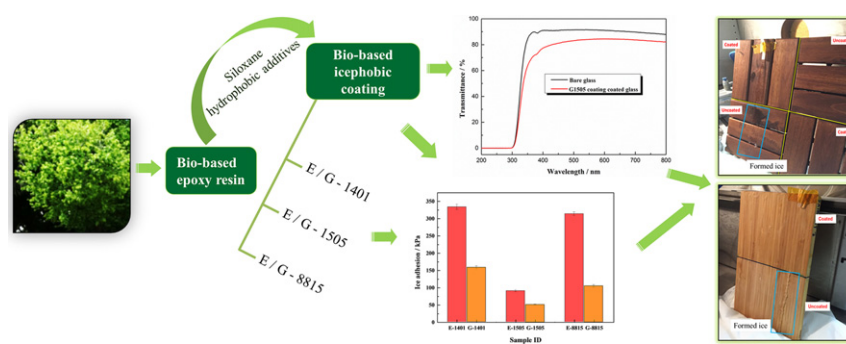
^a School of Materials Science and Engineering, Nanyang Technological University, 50 Nanyang Avenue, 639798, Singapore

^b Wolfson School of Mechanical Electrical and Manufacturing Engineering, Loughborough University, Loughborough, Leicestershire LE11 3TU, UK

HIGHLIGHTS

- Bio-based transparent icephobic coatings were formulated for room temperature processing.
- The addition of silanes enhanced the water repellency and reduces ice adhesion strength.
- The optimum coating exhibited ice adhesion around 50 kPa at -20°C and icing delay time of 2 min 45 s.
- As a demonstration, ice accumulation was much reduced on wooden floor plates and panels.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 13 July 2017

Received in revised form 6 October 2017

Accepted 7 December 2017

Available online 8 December 2017

Keywords:

Anti-icing
Icephobic
Transparent coating
Bio-epoxy
Ice adhesion
Icing delay
Water repellent

ABSTRACT

Ice accretion and accumulation pose serious challenges for maintaining the operation and performance of outdoor facilities in cold climate. Epoxy resin, with a wide range of formulation possibilities, is widely used as protective coatings for outdoor facilities. However, bisphenol A (BPA), a key ingredient of conventional epoxy, is known to interfere with human's natural hormones and cause various disorders in the body system. Reduction or complete elimination of the usage of BPA is therefore high in the agenda of the coatings industries. In this study, a transparent, anti-icing, bio-based epoxy coating was developed for room-temperature processing. As a result of hydrophobic treatment with addition of silanes, the glass-transition temperature and anti-icing performance of bio-based epoxy resin increased significantly. The optimum coating exhibited good water repellency and ice-adhesion strength as low as 50 kPa at -20°C , which was half of the widely accepted threshold value of 100 kPa for icephobic coatings. The icing delay time was much delayed compared with that of an uncoated glass substrate. To further demonstrate the anti-icing performance of the optimized coating, supercooled-water dripping on coated wooden outdoor floors and wooded boards was conducted at -15°C , superior anti-icing performance was observed on the coated substrates.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Icephobic coatings, as a passive strategy for anti-icing, are essential for a number of applications in cold climate or environment, including e.g. solar cells, windshields of automobiles, outdoor sports facilities,

marine and aerospace structures [1–3]. Earlier studies on icephobic coating were primarily focused on salt-glucose compounds [4], cellulose lacquers [5], liquid monolayers such as hydrocarbons, fluorocarbons, waxes, oil, as well as grease and rubber coatings [6–10]. A rapid development of superhydrophobic coatings in recent years [11] offers new opportunities for the design of icephobic coatings. A notion “icephobic coating” refers to several aspects of an icing event: i) a coating surface should enable water droplets to roll off easily or bounce away before

* Corresponding author.

E-mail addresses: wuxi0011@e.ntu.edu.sg (X. Wu), aszchen@ntu.edu.sg (Z. Chen).